PROPERTIES OF SOAP

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Soap is basically the sodium (sometimes potassium) salt of a long chain fatty acid. Figure 1 shows two representations of a soap.

1 What is meant by the lower case letter 'n' shown in the top diagram of figure 1?

Soap is a **surfactant** (surface active agent). This is a substance that, when added to a liquid, reduces its surface tension which increases its spreading and wetting properties. Surfactants must be able to deal with both water (polar) and lipid (non-polar) environments.

Soaps have a polar end (-COONa for the example shown in figure 2) and a non-polar end (a long hydrocarbon chain). The polar end ('head') is **hydrophilic** (literally 'loves water'; *hydro* meaning water, *phile* meaning loving) and it is water-soluble. The non-polar end ('tail') is **lipophilic** and is soluble in lipids. When soap is put into water, the structure breaks up into a positive (Na^+) ion and a negative (carboxylate) ion.

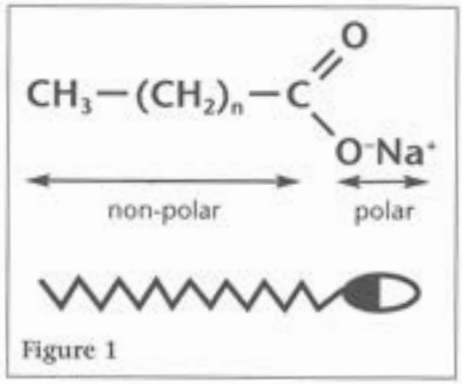


Figure 1

2 What do you think is the literal meaning of *lipophilic*?

3 Sometimes the term *hydrophobic* is used, literally meaning 'fear of water' (*phobos*, meaning fear). Give another term (already used on this page) that could be used to replace the term *hydrophobic* in reference to surfactants.

Surfactants have many uses. They:

- help dyes penetrate fabric evenly when textiles are dyed
- disperse aqueous suspensions of insoluble substances
- help produce aerosols
- aid in froth flotation purification of ores
- act as detergents.

A detergent is something that helps make dirty objects clean (*detergere*, meaning wipe clean). Often the 'dirt' is oil-based and cannot be efficiently removed by water alone. Soaps are detergents; they can dissolve their non-polar ends in lipids and their polar ends in water. The resulting complex is called a micelle (*mica*, grain or crumb). A micelle is a cluster of units, such as globules of grease or oil, surrounded by soap molecules. Detergents reorient their hydrophobic ends away from the water and their hydrophilic ends surround the 'dirt'; thus breaking it down and dissolving it.

4 Figure 2 shows how a soap (or similar detergent) can be used to clean grease from a plate. Refer to the diagrams to help you describe how soap does this. In your answer include the terms: *emulsion, hydrophilic, lipophilic, micelle, surfactant*.

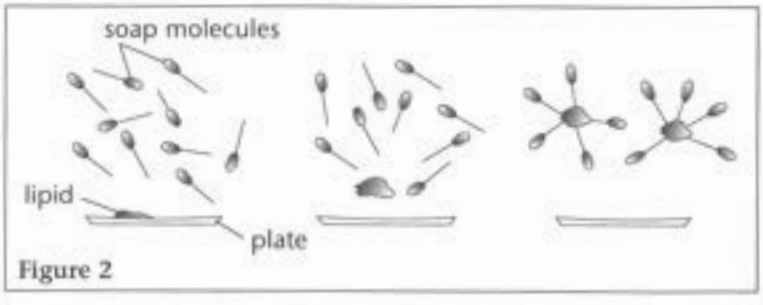


Figure 2

5 Explain why figure 2 is not drawn to scale.

6 Give three uses of surfactants, apart from acting as detergents.

The word *detergent* means a cleaning agent. Synthetic detergents (usually just called detergents) have similar properties to soap but are chemically different. Synthetic detergents do not form a scum with hard water.

7 What is meant by *synthetic*?

8 *A substance that increases the wetting and spreading properties of a liquid.* What does this define?

9 Define *scum*, in terms of chemistry.

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